

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			type of oil used, towel material and volume of hydrogen peroxide ☐ type of oil used, towel material and temperature of stain remover ☐ type of oil used and towel material ☒ type of oil used, towel material and cost of stain remover ☐			1	1		1
	(b)			it is the cheapest stain remover ☐ it is heat resistant ☐ it has a low concentration of hydrogen peroxide ☒ it takes a long time to work ☒ both needed for (1) 0 if more than two boxes ticked			1	1		1
	(c)			award (1) for either of following - stain remover D may have a catalyst added - stain remover D may have an enzyme added			1	1		1

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					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		0.14 / 0.143 (2) if answer incorrect award (1) for $\frac{0.4}{2.8}$ ecf possible for error in reading graph			2	2	2	
		(ii)		concentration S is half of concentration T (1) half the number of particles in the same volume (1) half the chance of successful collisions / half the number of successful collisions per second / half the frequency of successful collisions (1) award (2) if answered using 'lower' / 'fewer' rather than 'half' throughout	3			3		
				Question 8 total	3	0	5	8	2	3